Printed Pages : 06

TCS05027

Semester : V

Roll No. .....

B.Tech.

Examination, January-2023

(DESIGN & ANALYSIS OF ALGORITHM)

Maximum Marks : 70

Time Allowed : 3 Hours

Note: i) Attempt all the questions.

ii) Maximum marks assigned to each question is indicated.

Section - A

Q.1. Attempt all questions (2 marks questions) (2x5=10)

(a) Explain five main characteristics of an algorithm

(b) Discuss different types of rotations in RB trees.

(c) Differentiate between dynamic programming and greedy algorithm.

(d) Illustrate Depth First search with suitable example

(e) Write short note on NP-Complete problems.

Attemp all questions (Short answer questions)

(4x5=20)

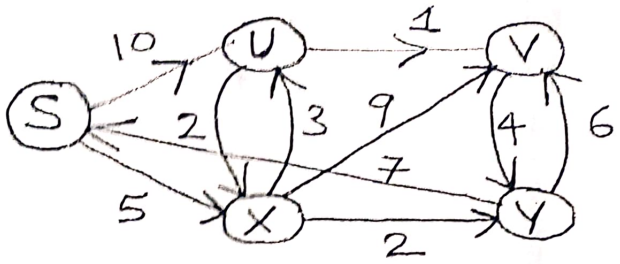
(f) Illustrate the operation of HEAP SORT on following array A = {6, 14, 3, 25, 2, 10, 20, 7, 6}

(g) Draw a binomial heap whose keys are element of A

A = {7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20}

(h) Explain in detail about 0/1 knapsack problem and fractional knapsock problem

(i) Find the shortest path in the following directed graph from the source vertex S to Y by using dijikstra algorithm



(j) Define naive string matching algorithm with suitable example.

Section - B

Q.2. Discuss the brief: best case, average case and worst case time complexity of analgorithm 8

OR

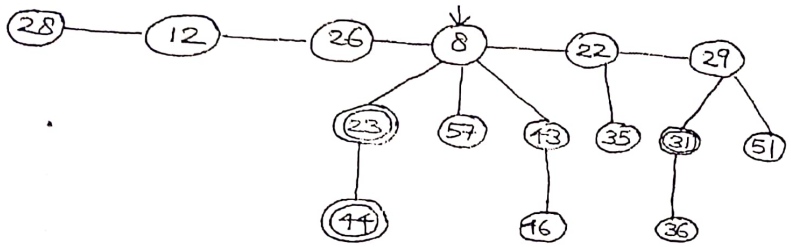
Solve the following recurrence relation

a) $T(n) = T(a,n) + T(1-a) + n, 0<a<1$

b) $T(n) = T(an/10) + n$

Q.3 Consider a Fibonoccl heap shown in below figure. Run the Fib-Heap extract-min and give the

resultant Fibonaccl Heap here double circle represent marked mode.



OR

Show the result of inserting the following items in an initally empty is tree of order 3

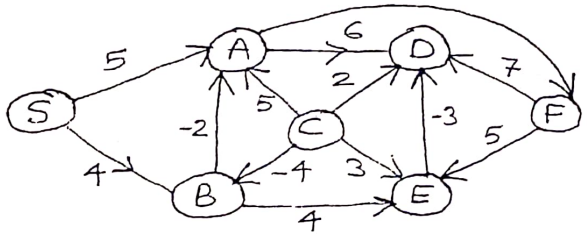
F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y D, Z, E, G, I

Q.4. Define Assembly Line scheduling problem and task scheduling problem. 8

OR

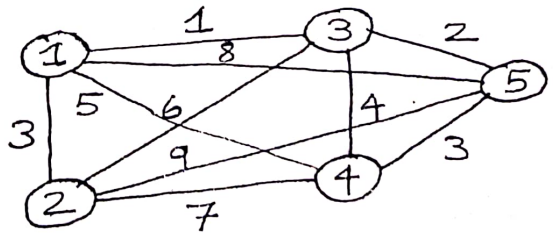
Discuss matrix chain multiplication with reference to dynamic programming technique whose sequence of dimension is <30, 1, 40, 10, 25>

Q.5. Find the shortest path using BELLMAN FORD algorithm form S to F in the following graph



OR

Solve the travelling salesman problem for the following graph by using branch and bound algorithm



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Q.6. Write short note on:

a) Knuth morris pratt algorithm for pattern matching

b) Randomized Sorting Algorithm

OR

Discuss P, NP-Complete problems, NP and NP-hard problems using suitable example